🎯 PROBLEM DEFINITION

Healthcare facilities often face **inefficient waiting management systems**, leading to **uncertainty, stress, and dissatisfaction among patients, caregivers, and staff**.

Patients experience **lack of real-time information**, unclear queue progression, and poor communication, which increases anxiety—especially in critical health situations. The waiting environment further amplifies discomfort due to overcrowding, noise, and absence of engaging or calming interventions.

Additionally, existing systems fail to address **diverse user needs**, including elderly users, children, and individuals with accessibility challenges. This results in an experience that is not only inefficient but also **emotionally overwhelming and inequitable**.

👉 **Core Problem Statement:**

"How might we design a smart, inclusive, and emotionally aware system that transforms hospital waiting into a transparent, stress-free, and engaging experience for all users?"

🔍 RESEARCH OVERVIEW

📊 1. Research Objective

To understand:

- User behavior during hospital waiting
- Emotional and psychological challenges
- Gaps in current hospital systems
- Opportunities for digital intervention

👥 2. Target Users Studied

- Patients (young, elderly, critical cases)
- Caregivers and family members
- Hospital staff
- First-time vs frequent hospital visitors

□ 3. Research Methods

a. User Personas

- 15 personas created representing diverse users
- Helped identify patterns in behavior and pain points

b. Empathy Mapping

- Captured what users **think, feel, see, hear, and do**
- Highlighted emotional stress and confusion

c. Problem Categorization

Problems grouped into:

- Information & transparency
- Emotional stress
- Accessibility barriers
- Communication gaps
- System inefficiencies

d. Feedback Simulation / Testing

- Collected feedback from 10 users
- Identified usability issues and strengths

📈 4. Key Insights

- **Uncertainty is the biggest stress trigger**
- Users value **predictability over speed**
- Emotional comfort is as important as functionality
- Accessibility is often ignored but critical
- Poor communication increases frustration more than delays

💡 5. Opportunity Areas

- Real-time queue tracking system
- AI-based wait-time prediction
- Calm/engagement features (distraction, relaxation)
- Voice + accessibility-first design
- Transparent communication system